Diego Bautista Avilés

PhD Candidate in Physics

diego.bau.avi@gmail.com

Research Interests

Non-Hermitian Quantum Mechanics · Topological Phases of Matter · Floquet Systems · Chiral Phonons

Education

Aug 2024–present **Doctor of Philosophy in Physics**, *Universidad de Chile*, Santiago, Chile
PhD Candidate (thesis proposal defended and approved). Supervisor: Prof. Luis E. F. Foà Torres. Doctoral Fellow, ANID. *Thesis*: *Topological Phases Induced by Exceptional Steering Protocols*. I engineer non-Hermitian topological phases in Floquet lattice models through protocols that alternate Hermitian and non-Hermitian dynamics (analogous to successive quantum measurements), targeting photonic-network emulators as experimental platforms.

2019–2023 **Bachelor of Science in Physics**, *Universidad Nacional Autónoma de México (UNAM)*, Mexico City, Mexico
GPA 9.95/10, ranked 1st out of 422 students. **Gabino Barreda Medal (UNAM)**: highest GPA of the Physics graduating class.

2016–2019 **High School (International Baccalaureate, graduated with honours)**, *Tecnológico de Monterrey (ITESM)*, Mexico

Research & Teaching Experience

Aug 2024–present **Teaching Assistant**, *Universidad de Chile, Faculty of Physical and Mathematical Sciences*, Santiago, Chile
Courses: *Quantum Mechanics I* (Aug–Dec 2024; Aug–Dec 2025; Mar 2026–present), *Classical Mechanics* (Mar–May 2025), *Quantum Mechanics II* (Aug–Dec 2025; Mar 2026–present).

Jun–Jul 2025 **Simons Visitor**, *International Centre for Theoretical Physics (ICTP)*, Trieste, Italy
I worked with Prof. Luis E. F. Foà Torres (Universidad de Chile).

Jan–Feb 2025 **Research Visit**, *Karlsruhe Institute of Technology (KIT)*, Karlsruhe, Germany
I worked in the group of Prof. Alexander Mirlin.

Jun–Jul 2024 **Simons Visitor**, *International Centre for Theoretical Physics (ICTP)*, Trieste, Italy
I worked with Prof. Luis E. F. Foà Torres (Universidad de Chile).

Aug–Dec 2023 **Teaching Assistant, Atomic Physics & Condensed Matter**, *UNAM, Faculty of Sciences*, Mexico City, Mexico

Jun–Jul 2023 **Summer Research Internship**, *University of Vienna*, Vienna, Austria
I was awarded a research contract as part of the Vienna Doctoral School of Physics program. Working under the supervision of Prof. Dr. Toma Susi (Physics of Nanostructured Materials Group), I extended the open-source abTEM STEM-simulation library by integrating DFT-derived (GPAW) inter-ionic electrostatic potentials to account for bonding effects in ionic materials neglected by the existing implementation, and designed systematic benchmarks evaluating their impact on simulated images.

Mar 2021–May 2024 **Research Fellow**, *UNAM, Faculty of Chemistry*, Mexico City, Mexico
I worked under the supervision of Prof. Dr. José Eduardo Barrios Vargas, in collaboration with Prof. Luis E. F. Foà Torres (Universidad de Chile). I developed numerical codes to model phonon chirality in carbon nanotubes and helical structures, implementing pseudoangular momentum and Hausdorff chirality measures. I identified C3-symmetric structures as optimal platforms for chiral phonons and ruled out carbon nanotubes as candidates, presenting results at multiple national and international conferences. *Dec 2023–May 2024: additionally affiliated with the CONAHCYT Frontier Research Project.*

Publications

[1] G. Pappas, **D. Bautista Avilés**, L. E. F. Foà Torres, and V. Achilleos, "Critical timescales for chiral state conversion and irreversibility," *Phys. Rev. A*, vol. 114, no. 4, p. 042217, 2026. DOI: 10.1103/5w1p-sg68

[2] L. E. F. Foà Torres, G. Pappas, V. Achilleos, and **D. Bautista Avilés**, "Precision-induced irreversibility in non-Hermitian systems," *Phys. Rev. A*, vol. 114, no. 1, p. 012207, 2026. DOI: 10.1103/mrmj-59p6

Skills

Languages Spanish (Native), English (Advanced, TOEFL certified)

Programming Python, Mathematica, LaTeX, Java

Schools & Conferences

Oct 2026 **Joint ICTP-WE Heraeus College on Advances in Topological Quantum Matter**, *ICTP*, Trieste, Italy, Poster
Non-Hermitian Anomalous Floquet-Anderson Insulator.

Jul 2026 **XXVIII Giambiagi Winter School "New Directions in Condensed Matter"**, Buenos Aires, Argentina, Poster
Non-Hermitian Anomalous Floquet-Anderson Insulator.

Dec 2025 **X Reunión de Sólidos**, Bariloche, Argentina, Poster
Non-Hermitian Anomalous Floquet-Anderson Insulator.

Nov 2025 **QUOST IX**, Valparaíso, Chile, Poster
Non-Hermitian Anomalous Floquet-Anderson Insulator.

Sep–Oct 2025 **XXIII B-RMS Meeting**, Salvador, Brazil, Poster
Exploring Chiral Phonons in Simple Lattice Models.

May 2025 **Joint ICTP-Lincei Conference on Quantum Physics: from Foundations to Emerging Technologies**, *ICTP*, Trieste, Italy, Attendee

Nov 2023 **Gathering on Transport at Nanoscale**, Cuernavaca, Mexico, Oral presentation
Phonon Pseudo Angular Momentum in Graphene Structures.

Oct 2023 **Congreso Nacional de Física**, Morelia, Mexico, Poster
Phonon Pseudo Angular Momentum in Graphene Structures.

Mar 2023 **Conference and Advanced School on Low-Dimensional Quantum Systems (ICTP)**, Santiago, Chile, Attendee

Nov 2022 **Reunión Mexicana de Físico-Química Teórica**, Cuernavaca, Mexico, Poster
Directional Control of Electric Current in a Ring-Shaped Device Partially Covered with a Graphene Bilayer.

Awards & Honours

2023 **Gabino Barreda Medal**, *UNAM*
Highest GPA of the Physics graduating class.

2016–2019 **High Academic Achievement Scholarship**, *ITESM*

2018 **Silver Medal**, *Mathematics Olympiad, Mexico City*

References

Prof. Luis E. F. Foà Torres
Professor
Department of Physics, Universidad de Chile
luis.foatorres@uchile.cl

Prof. José Eduardo Barrios
Associate Professor
Dept. of Physical and Theoretical Chemistry, UNAM
j.e.barrios@gmail.com